Week 12 Reading

Multi-Objective Optimization

Draft Material — This content is under development and subject to change.

1 Overview

S. Yang, L. Ruangpan, A. S. Torres, and Z. Vojinovic [1]

2 Discussion Questions

1. Yang et al. optimize for both flood reduction benefits and ecosystem co-benefits of nature-based solutions. Why do the authors treat these as separate objectives? What information would a weighted sum lose?
2. The paper uses NSGA-II to approximate the Pareto front. What role does the choice of algorithm play in the results? How would you assess whether the algorithm has converged to a good approximation of the true front?
3. Look at the Pareto front figures in the paper. Pick two solutions on opposite ends of the front and describe the trade-off a decision-maker would face in choosing between them. Is there a "knee" — a region where small sacrifices on one objective yield large gains on the other?
4. The authors apply their framework to Sint Maarten, a small island recovering from Hurricane Irma. How does the local context (limited resources, post-disaster recovery, small geographic scale) shape which objectives matter and how results should be communicated to decision-makers?
5. Compare this paper's approach to the benefit-cost analysis framework from Week 6. Under what conditions is BCA sufficient? What does multi-objective optimization provide that BCA does not?

Bibliography

[1] S. Yang, L. Ruangpan, A. S. Torres, and Z. Vojinovic, "Multi-Objective Optimisation Framework for Assessment of Trade-Offs between Benefits and Co-benefits of Nature-based Solutions," *Water Resources Management*, vol. 37, no. 6–7, pp. 2325–2345, May 2023, doi: 10.1007/s11269-023-03470-8.